Siyuan Lu

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Education

Zhejiang University / *Computer Science and Technology* / *Ph.D.* 2025 – 2030 (Expected)

Joint program with Westlake University and Shanghai Innovation Institute
Research: Agentic Systems and Agent Training Advisors: *Tao Lin*

Beijing University of Posts and Telecommunications / *Internet of Things* / *B.E.* 2021 – 2025

Joint program with Queen Mary University of London
Weighted Average Mark: 90.97/100 Rank: 7/182

Published Works

Don't Just Fine-tune the Agent, Tune the Environment

First Author ICLR 2026

Proposed Environment Tuning paradigm that enables agents to learn complex behaviors directly from problem instances without pre-collected expert trajectories. Through structured curriculum, environment augmentation, and fine-grained rewards, achieved superior tool-calling capabilities with only 400 samples.

Teach the Magnitude, Not the Direction: Verifier-Anchored Credit Assignment for Multi-Turn Multi-Step LLM Agents

Co-author EMNLP 2026 Main

Proposed CREST, a hierarchical credit-assignment framework for multi-turn, multi-step tool-use agents that combines turn-segmented verifiable rewards with entropy-gated self-teacher signals for token-level optimization. CREST preserves a verifier-bounded performance ceiling and consistently outperforms RL and distillation baselines on BFCL V3 and WildToolBench.

AWorld: Orchestrating the Training Recipe for Agentic AI

Co-author Technical Report

Proposed an open-source system for large-scale agent-environment interaction, achieving 14.6x speedup in experience collection through distributed deployment. The trained Qwen3-32B model reached 32.23% accuracy on GAIA test set, surpassing GPT-4o.

PreAct: Predicting Future in ReAct Enhances Agent's Planning Ability

Third Author COLING 2025

Proposed PreAct framework that integrates prediction, reasoning, and action. By leveraging predictive information, enables agents to provide broader and more strategic reasoning, outperforming ReAct methods on complex tasks.

Work Experience

Tencent Hunyuan 2026.9 – Present

Research Intern, Project UP T3 Agent Teams, Hunyuan Foundation Model

Mentor: *Wei Shen*
Participated in the development and release of the Hy4 foundation model, working on reinforcement learning for coding agents and the harness-based agentic task RL big run covering 15 task categories.

Ant Group 2025.6 – 2026.6

Research Intern AWorld Team, InclusionAI

Mentor: *Chenyi Zhuang*

Participated in research and development of the AWorld project, producing ICLR and EMNLP conference papers on post-training of large models in the tool-calling domain.

Open Source Contributions

Hy4 770B-A49B MoE

Contributor Foundation Model, Vision, Computer Use

Hy4 is Tencent's latest-generation large language model, released with open weights in November 2026 and placing Hy among the strongest open-source models. It is the first in the series to add vision understanding and computer use, combining perception, reasoning and tool use for fine-grained image and chart analysis and operating across CLI and GUI on long-horizon cross-application workflows. Main contributions: Participated in RL training for agentic capabilities, including coding agents and multi-category agentic tasks.

801-30B 30B-A3B MoE

Contributor Pre-training, Recursive Self-Improvement, Agentic Data

801-30B-A3B-Base is the first public work applying recursive self-improvement (RSI) to a full frontier pre-training run; the 30B-A3B MoE outperforms Qwen3-30B-A3B as a base model and, with only lightweight SFT, surpasses fully post-trained baselines on agentic benchmarks. Main contributions: Responsible for environment synthesis and validation in the agentic data pipeline, where agents turn real GitHub PR-issues into reproducible, parallel-executable environments, with layered rules blocking shortcuts and answer leakage; the pipeline produced 220K environments and 1.1M verified trajectories (100B+ tokens) for pre-training.

AWorld 1.2k Stars

Contributor Agentic AI, Self-Evolution, Multi-Agent Systems

AWorld is a self-evolving agent framework providing a complete agent learning loop (environment access, agent construction, experience retrieval, model training), with multi-agent collaboration, rich tool integration, advanced context management and an event-driven architecture. Main contributions: Participated in designing the training module, enabling training frameworks (such as verl) to leverage AWorld's agent abstraction and distributed environment deployment capabilities, accelerating rollout for large model RL training.

Honors

First-Class Scholarship, Beijing University of Posts and Telecommunications 2023

Second-Class Scholarship, Beijing University of Posts and Telecommunications 2022

Skills

IELTS 7.0, proficient in English listening, speaking, reading, and writing

Daily development on Linux; proficient with Git, Docker and GitHub for version control

Modern Python development, proficient with the latest toolchain such as uv and ruff

Inference engines: sglang, vllm; LLM reinforcement learning frameworks: verl, slime, miles